MAPS FROM A FILTERED ACYCLIC COMPLEX INTO A BOUNDED BELOW COMPLEX OF FILTERED INJECTIVES ARE NULL HOMOTOPIC

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 8423-8455, following the statement at 8413-8421.

Editorial note. No omissions: the induction step is displayed with the exact sequence and the lifting lemma.

Allowed standard prerequisites (editorial).

- Structure of filtered acyclic complexes. Homology, Lemma homology-lemma-filtered-acyclic (cross-chapter, cited at ~8445). Strictness of the differentials and the exact sequence 0 -> Im(d^{n-1}) -> K^n -> Im(d^n) -> 0 in Fil^f(A).
- Lifting against filtered injectives. lemma-injective-property-filtered-injective (same chapter, cited at ~8455). Extends the map along the strict monomorphism Im(d^n) -> K^{n+1} to h: K^{n+1} -> I^n.

Source reference. Original reference 05QH: focused-seeds/geometry/S9-der-filtered-acyclic-is-zero/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05TR: sources/stacks-derived/pinned-original-source.tex, source lines 8113–8113. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $K^\bullet, I^\bullet \in K(Fil^f(\mathcal{A}))$. Assume $K^\bullet$ is filtered acyclic and $I^\bullet$ bounded below and consisting of filtered injective objects. Any morphism $K^\bullet \to I^\bullet$ is homotopic to zero:* $\mathrm{Hom}_{K(Fil^f(\mathcal{A}))}(K^\bullet, I^\bullet) = 0$.

Proof. Let $\alpha : K^\bullet \to I^\bullet$ be a morphism of complexes. Assume that $\alpha^j = 0$ for $j < n$. We will show that there exists a morphism $h : K^{n+1} \to I^n$ such that $\alpha^n = h \circ d$. Thus α will be homotopic to the morphism of complexes β defined by

$$\beta^j = \begin{cases} 0 & \text{if} \quad j \leq n \\ \alpha^{n+1} - d \circ h & \text{if} \quad j = n+1 \\ \alpha^j & \text{if} \quad j > n+1 \end{cases}$$

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This will clearly prove the lemma (by induction). To prove the existence of h note that $\alpha^n \circ d_K^{n-1} = 0$ since $\alpha^{n-1} = 0$. Since $K^\bullet$ is filtered acyclic we see that d_K^{n-1} and d_K^n are strict and that

$$0 \to \mathrm{Im}(d_K^{n-1}) \to K^n \to \mathrm{Im}(d_K^n) \to 0$$

is an exact sequence of the exact category $\mathrm{Fil}^f(\mathcal{A})$, see Homology, Lemma 05QH. Hence we can think of α^n as a map into I^n defined on $\mathrm{Im}(d_K^n)$. Using that $\mathrm{Im}(d_K^n) \to K^{n+1}$ is a strict monomorphism and that I^n is filtered injective we may lift this map to a map $h : K^{n+1} \to I^n$ as desired, see Lemma 05TR. $\square$